Grade 10 – Term 1
C1–C7 Catch-up Exam Solutions – Version 2

2026–2027

Contents

1 Community Science Sessions [C1-3 C2-6] 3

2 Kayak Rentals [C1-3 C3-6] 4

3 Bicycle Assembly [C4-2] 5

4 Solar Installation [C4-2] 6

5 Equipment Leasing [C4-2] 7

6 Festival Meals [C5-2] 8

7 Laundry Capacity [C5-2] 9

8 Juice Production [C5-2] 10

9 Coffee Roasting [C6-2] 11

10 Farm Visitor Facilities [C6-2] 12

11 Translation Capacity [C6-2] 13

12 Mobile Game Release [C7-3] 14

13 Timber Supply Agreement [C7-3] 15

14 Summarized Solutions [C1–C7] 16

Learning objective

Characterise economic decisions under uncertainty, organise payoff tables, calculate profits from revenues and costs, and justify choices using Maximax, Maximin, Minimax Regret, EMV/Bayes, and probability sensitivity analysis.

Criteria assessment

This exam assesses C1–C7. Each question and task shows its points. **Passing requires a minimum of 5 cumulative points per criterion, including points earned in previous learning activities**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

C5 Constructs opportunity loss tables and applies the Minimax Regret criterion to select the alternative that minimises the maximum opportunity loss.

C6 Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.

C7 Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

№	C1	C2	C3	C4	№	C5	C6	C7
1	3	6	–	–	6	2	–	–
2	3	–	6	–	7	2	–	–
3	–	–	–	2	8	2	–	–
4	–	–	–	2	9	–	2	–
5	–	–	–	2	10	–	2	–
					11	–	2	–
					12	–	–	3
					13	–	–	3
Tot.	**6**	**6**	**6**	**6**	**Tot.**	**6**	**6**	**6**

Each criterion has 6 available points; the exam has 42 available points overall. **Notation.** A_i and S_j always follow the original alternative and state order within each problem. $P(A_i, S_j)$ denotes a monetary payoff; $P(S_j)$ denotes a probability. Navigation arrows return to the contents, the problem, or the summary.

1 Community Science Sessions [C1-3 C2-6]

A community science centre must choose an astronomy session or a robotics session before family demand is known. Demand will be busy or quiet. Astronomy provides a net contribution of 14 USD per participant and a fixed sponsorship contribution of 90 USD per session. Robotics provides 17 USD per participant and a fixed sponsorship contribution of 65 USD per session. Busy demand adds a further net contribution of 5 USD per participant to either session; quiet demand adds 3 USD per participant. Astronomy attracts 28 participants with busy demand or 17 with quiet demand; robotics attracts 23 or 14, respectively. All contributions are additive, and no other adjustments are needed.

Questions/Tasks

C1 Characterize the key elements of the decision-making situation **[C1-3]**

C2 and construct the payoff table. **[C2-6]**

C1 - Characterization [3]

Element	Description	
Alternatives	**A1:** Astronomy session	**A2:** Robotics session
States of nature	**S1:** Busy family demand	**S2:** Quiet family demand
Events	Family demand revealed after the session is chosen; the centre cannot control it.	
Consequences	Total net contribution in USD for each session–demand pair.	

The economic context is a community science centre choosing its session before family demand is known. The decision controls the session, while demand controls attendance and the extra contribution rate.

C2 - Additive contributions and payoff table [6]

Let q_{ij} be participants, v_i the session contribution per participant, d_j the demand contribution per participant, and f_i the fixed sponsorship. All monetary rates are in USD per participant; f_i and payoffs are in USD.

$$P(A_i, S_j) = q_{ij}(v_i + d_j) + f_i.$$

The per-participant contributions are added before multiplying by participants. The fixed sponsorship is then added once per session; it is not multiplied by participants.

$$\begin{aligned} P(A_1, S_1) &= 28(14 + 5) + 90 = 532 + 90 = 622, \\ P(A_1, S_2) &= 17(14 + 3) + 90 = 289 + 90 = 379, \\ P(A_2, S_1) &= 23(17 + 5) + 65 = 506 + 65 = 571, \\ P(A_2, S_2) &= 14(17 + 3) + 65 = 280 + 65 = 345. \end{aligned}$$

Completed payoff table (USD).

Alternative	Busy (S_1)	Quiet (S_2)
Astronomy session (A_1)	622	379
Robotics session (A_2)	571	345

Conclusion: the astronomy session gives a higher contribution in both states: USD 622 under busy demand and USD 379 under quiet demand.

2 Kayak Rentals [C1-3 C3-6]

A kayak rental business must choose single kayaks or tandem kayaks before seasonal demand is known. Demand will be busy or quiet. Single kayaks generate 84 rentals with busy demand or 56 with quiet demand; tandem kayaks generate 66 or 38, respectively. Base revenue per rental is 36 USD for the single kayaks and 49 USD for the tandem kayaks. Busy demand adds a surcharge of 7 USD per rental to either rental type; quiet demand adds 4 USD per rental. Variable cost per rental is 20 USD for the single kayaks and 29 USD for the tandem kayaks. Total fixed seasonal costs, for either rental type, are 540 USD with busy demand and 620 USD with quiet demand. These fixed costs are paid once per season, and there are no other revenues or costs.

Questions/Tasks

C1 Characterize the key elements of the decision-making situation **[C1-3]**
C3 and construct the payoff table. **[C3-6]**

C1 - Characterization [3]

Element	Description	
Alternatives	**A1:** Single kayaks	**A2:** Tandem kayaks
States of nature	**S1:** Busy seasonal demand	**S2:** Quiet seasonal demand
Events	Seasonal demand revealed after the rental type is chosen; the operator cannot control it.	
Consequences	Seasonal profit in USD for each rental type–demand pair.	

The economic context is a kayak rental business choosing a rental type before knowing sales, the demand surcharge, and the seasonal fixed cost.

C3 - Profit calculations and payoff table [6]

Profit equals total revenue minus total variable costs minus fixed seasonal costs:

$$R_{ij} = q_{ij}(b_i + s_j), \qquad VC_{ij} = q_{ij}c_i,$$
$$P(A_i, S_j) = R_{ij} - VC_{ij} - F_j = q_{ij}(b_i + s_j) - q_{ij}c_i - F_j.$$

Here q_{ij} is the number of rentals sold; b_i, s_j, and c_i are base revenue, surcharge, and variable cost in USD per rental; F_j is the fixed seasonal cost in USD. The surcharge is part of revenue for every rental sold. The fixed cost is paid once in each alternative–state calculation, not once per rental.

$$P(A_1, S_1) = 84(36 + 7) - 84(20) - 540 = 3612 - 1680 - 540 = 1392,$$
$$P(A_1, S_2) = 56(36 + 4) - 56(20) - 620 = 2240 - 1120 - 620 = 500,$$
$$P(A_2, S_1) = 66(49 + 7) - 66(29) - 540 = 3696 - 1914 - 540 = 1242,$$
$$P(A_2, S_2) = 38(49 + 4) - 38(29) - 620 = 2014 - 1102 - 620 = 292.$$

Completed payoff table (profit in USD).

Alternative	Busy (S_1)	Quiet (S_2)
Single kayaks (A_1)	1392	500
Tandem kayaks (A_2)	1242	292

Conclusion: single kayaks earn more in both states: USD 1,392 under busy demand and USD 500 under quiet demand.

3 Bicycle Assembly [C4-2]

A bicycle manufacturer must select an assembly plan before market conditions are known. Licensed production earns a fixed royalty regardless of the market state. The possible states, in column order, are Boom, Stable market, Recession. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Large factory (A_1)	132	53	-27
Standard line (A_2)	91	63	19
Compact line (A_3)	61	47	33
Licensed production (A_4)	26	26	26

Questions/Tasks

C4 What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

C4 - Maximax and Maximin [2]

Alternatives: A_1: Large factory; A_2: Standard line; A_3: Compact line; A_4: Licensed production.

States: S_1: Boom; S_2: Stable market; S_3: Recession. All profits are in thousand USD. Find the largest and smallest payoff in every row:

Alternative	Row maximum M_i	Row minimum m_i
Large factory (A_1)	$\max\{132, 53, -27\} = 132$	$\min\{132, 53, -27\} = -27$
Standard line (A_2)	$\max\{91, 63, 19\} = 91$	$\min\{91, 63, 19\} = 19$
Compact line (A_3)	$\max\{61, 47, 33\} = 61$	$\min\{61, 47, 33\} = 33$
Licensed production (A_4)	$\max\{26, 26, 26\} = 26$	$\min\{26, 26, 26\} = 26$

$$\max_i M_i = \max\{132, 91, 61, 26\} = 132, \qquad \max_i m_i = \max\{-27, 19, 33, 26\} = 33.$$

Maximax: choose **Large factory** (A_1), with best-case profit 132 thousand USD.

Maximin: choose **Compact line** (A_3), with worst-case profit 33 thousand USD. Maximax focuses on the best possible profit and ignores the downside. Maximin protects the best worst-case profit and may forgo a large gain. Without probabilities, neither criterion estimates likelihoods or expected profit.

4 Solar Installation [C4-2]

A solar installer must select an installation programme before customer adoption is known. The service agreement pays a fixed profit under every adoption state. The possible states, in column order, are Rapid adoption, Moderate adoption, Slow adoption. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Regional rollout (A_1)	168	51	-39
Town rollout (A_2)	108	69	17
Village rollout (A_3)	73	52	31
Service agreement (A_4)	35	35	35

Questions/Tasks

C4 What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

C4 - Maximax and Maximin [2]

Alternatives: A_1: Regional rollout; A_2: Town rollout; A_3: Village rollout; A_4: Service agreement.

States: S_1: Rapid adoption; S_2: Moderate adoption; S_3: Slow adoption. All profits are in thousand USD. Find the largest and smallest payoff in every row:

Alternative	Row maximum M_i	Row minimum m_i
Regional rollout (A_1)	$\max\{168, 51, -39\} = 168$	$\min\{168, 51, -39\} = -39$
Town rollout (A_2)	$\max\{108, 69, 17\} = 108$	$\min\{108, 69, 17\} = 17$
Village rollout (A_3)	$\max\{73, 52, 31\} = 73$	$\min\{73, 52, 31\} = 31$
Service agreement (A_4)	$\max\{35, 35, 35\} = 35$	$\min\{35, 35, 35\} = 35$

$$\max_i M_i = \max\{168, 108, 73, 35\} = 168, \qquad \max_i m_i = \max\{-39, 17, 31, 35\} = 35.$$

Maximax: choose **Regional rollout** (A_1), with best-case profit 168 thousand USD.

Maximin: choose **Service agreement** (A_4), with worst-case profit 35 thousand USD. Maximax focuses on the best possible profit and ignores the downside. Maximin protects the best worst-case profit and may forgo a large gain. Without probabilities, neither criterion estimates likelihoods or expected profit.

5 Equipment Leasing [C4-2]

A construction-equipment business must choose a fleet plan before next season's rental utilisation is known. The possible states, in column order, are Peak utilisation, Busy utilisation, Routine utilisation, Quiet utilisation. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3	S_4
Buy large fleet (A_1)	156	81	21	-46
Buy medium fleet (A_2)	113	85	41	11
Lease small fleet (A_3)	77	63	49	25
Pool equipment (A_4)	54	51	39	32

Questions/Tasks

C4 What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

C4 - Maximax and Maximin [2]

Alternatives: A_1: Buy large fleet; A_2: Buy medium fleet; A_3: Lease small fleet; A_4: Pool equipment.

States: S_1: Peak utilisation; S_2: Busy utilisation; S_3: Routine utilisation; S_4: Quiet utilisation. All profits are in thousand USD. Find the largest and smallest payoff in every row:

Alternative	Row maximum M_i	Row minimum m_i
Buy large fleet (A_1)	$\max\{156, 81, 21, -46\} = 156$	$\min\{156, 81, 21, -46\} = -46$
Buy medium fleet (A_2)	$\max\{113, 85, 41, 11\} = 113$	$\min\{113, 85, 41, 11\} = 11$
Lease small fleet (A_3)	$\max\{77, 63, 49, 25\} = 77$	$\min\{77, 63, 49, 25\} = 25$
Pool equipment (A_4)	$\max\{54, 51, 39, 32\} = 54$	$\min\{54, 51, 39, 32\} = 32$

$$\max_i M_i = \max\{156, 113, 77, 54\} = 156, \qquad \max_i m_i = \max\{-46, 11, 25, 32\} = 32.$$

Maximax: choose **Buy large fleet** (A_1), with best-case profit 156 thousand USD.

Maximin: choose **Pool equipment** (A_4), with worst-case profit 32 thousand USD. Maximax focuses on the best possible profit and ignores the downside. Maximin protects the best worst-case profit and may forgo a large gain. Without probabilities, neither criterion estimates likelihoods or expected profit.

6 Festival Meals [C5-2]

A festival caterer must select a meal-preparation plan before attendance is known. The possible states, in column order, are Crowded, Average attendance, Sparse attendance. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Full menu (A_1)	87	37	-16
Balanced menu (A_2)	65	54	21
Limited menu (A_3)	43	41	34

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Minimax Regret [2]

Alternatives: A_1: Full menu; A_2: Balanced menu; A_3: Limited menu.

States: S_1: Crowded; S_2: Average attendance; S_3: Sparse attendance.

$$\boldsymbol{b} = (87, 54, 34), \qquad b_j = \max_k P(A_k, S_j).$$

Each column is compared with its highest profit. Regret is the profit missed by not choosing that state's best alternative:

$$r(A_i, S_j) = \max_k P(A_k, S_j) - P(A_i, S_j).$$

All payoffs and regrets in this calculation are in thousand USD.

Alternative	S_1	S_2	S_3	Maximum regret
A_1	$87-87$ $=0$	$54-37$ $=17$	$34-(-16)$ $=50$	50
A_2	$87-65$ $=22$	$54-54$ $=0$	$34-21$ $=13$	22
A_3	$87-43$ $=44$	$54-41$ $=13$	$34-34$ $=0$	44

$$\min_i \max_j r(A_i, S_j) = \min\{50, 22, 44\} = 22.$$

Choose **Balanced menu** (A_2): its maximum regret is **22 thousand USD**. This is the smallest worst possible opportunity loss; it does not mean that this loss occurs in every state or that this alternative always earns the most profit.

7 Laundry Capacity [C5-2]

A commercial laundry must select its capacity plan before hotel orders are known. A contracted service provides a fixed profit under every workload state. The possible states, in column order, are Heavy workload, Routine workload, Light workload. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Build plant (A_1)	126	47	-32
Upgrade machines (A_2)	94	73	19
Lease machines (A_3)	69	61	38
Contract service (A_4)	41	41	41

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Minimax Regret [2]

Alternatives: A_1: Build plant; A_2: Upgrade machines; A_3: Lease machines; A_4: Contract service.

States: S_1: Heavy workload; S_2: Routine workload; S_3: Light workload.

$$\boldsymbol{b} = \big(126, 73, 41\big), \qquad b_j = \max_k P(A_k, S_j).$$

Each column is compared with its highest profit. Regret is the profit missed by not choosing that state's best alternative:

$$r(A_i, S_j) = \max_k P(A_k, S_j) - P(A_i, S_j).$$

All payoffs and regrets in this calculation are in thousand USD.

Alternative	S_1	S_2	S_3	Maximum regret
A_1	$126-126$ $=0$	$73-47$ $=26$	$41-(-32)$ $=73$	73
A_2	$126-94$ $=32$	$73-73$ $=0$	$41-19$ $=22$	32
A_3	$126-69$ $=57$	$73-61$ $=12$	$41-38$ $=3$	57
A_4	$126-41$ $=85$	$73-41$ $=32$	$41-41$ $=0$	85

$$\min_i \max_j r(A_i, S_j) = \min\{73, 32, 57, 85\} = 32.$$

Choose **Upgrade machines** (A_2): its maximum regret is **32 thousand USD**. This is the smallest worst possible opportunity loss; it does not mean that this loss occurs in every state or that this alternative always earns the most profit.

8 Juice Production [C5-2]

A juice producer must choose its daily production plan before sales are known. The possible states, in column order, are Exceptional sales, Brisk sales, Ordinary sales, Poor sales. Profits are in USD. No probabilities are available.

Alternative	S_1	S_2	S_3	S_4
Large batch (A_1)	1080	610	230	-270
Staged batches (A_2)	810	730	510	150
Small batch (A_3)	530	550	480	390

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Minimax Regret [2]

Alternatives: A_1: Large batch; A_2: Staged batches; A_3: Small batch.

States: S_1: Exceptional sales; S_2: Brisk sales; S_3: Ordinary sales; S_4: Poor sales.

$$\boldsymbol{b} = \big(1080, 730, 510, 390\big), \qquad b_j = \max_k P(A_k, S_j).$$

Each column is compared with its highest profit. Regret is the profit missed by not choosing that state's best alternative:

$$r(A_i, S_j) = \max_k P(A_k, S_j) - P(A_i, S_j).$$

All payoffs and regrets in this calculation are in USD.

Alternative	S_1	S_2	S_3	S_4	Maximum regret
A_1	$1080-1080$ $=0$	$730-610$ $=120$	$510-230$ $=280$	$390-(-270)$ $=660$	660
A_2	$1080-810$ $=270$	$730-730$ $=0$	$510-510$ $=0$	$390-150$ $=240$	270
A_3	$1080-530$ $=550$	$730-550$ $=180$	$510-480$ $=30$	$390-390$ $=0$	550

$$\min_i \max_j r(A_i, S_j) = \min\{660, 270, 550\} = 270.$$

Choose **Staged batches** (A_2): its maximum regret is **270 USD**. This is the smallest worst possible opportunity loss; it does not mean that this loss occurs in every state or that this alternative always earns the most profit.

9 Coffee Roasting [C6-2]

A coffee roaster must choose a roasting plan before export conditions are known. The possible states, in column order, are Export boom, Regular exports, Export slump. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$	0.35	0.40	0.25
Specialty roast (A_1)	98	49	-17
Blended roast (A_2)	74	63	27
Everyday roast (A_3)	51	53	42

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - EMV / Bayes [2]

Alternatives: A_1: Specialty roast; A_2: Blended roast; A_3: Everyday roast.

States: S_1: Export boom; S_2: Regular exports; S_3: Export slump.

$$\boldsymbol{q} = (0.35, 0.40, 0.25), \qquad 0.35 + 0.40 + 0.25 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P(A_i, S_j).$$

Multiply each state payoff by its probability, including any negative payoff, and add the weighted amounts. All EMVs are in thousand USD.

$$\begin{aligned} EMV(A_1) &= 0.35(98) + 0.40(49) + 0.25(-17) \\ &= 34.30 + 19.60 - 4.25 = 49.65, \\ EMV(A_2) &= 0.35(74) + 0.40(63) + 0.25(27) \\ &= 25.90 + 25.20 + 6.75 = 57.85, \\ EMV(A_3) &= 0.35(51) + 0.40(53) + 0.25(42) \\ &= 17.85 + 21.20 + 10.50 = 49.55. \end{aligned}$$

$$\max\{49.65, 57.85, 49.55\} = 57.85.$$

Choose **Blended roast** (A_2), with expected profit **57.85 thousand USD** (USD 57,850). EMV is a probability-weighted expected payoff, not a guaranteed outcome. The realised profit will be the payoff for the state that actually occurs.

10 Farm Visitor Facilities [C6-2]

A farm must select visitor facilities before seasonal attendance is known. Leasing to an operator earns a fixed profit under all attendance states. The possible states, in column order, are Busy season, Typical season, Quiet season. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$	0.25	0.45	0.30
Visitor centre (A_1)	117	43	-29
Covered terrace (A_2)	89	64	23
Picnic shelter (A_3)	63	57	40
Lease to operator (A_4)	32	32	32

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - EMV / Bayes [2]

Alternatives: A_1: Visitor centre; A_2: Covered terrace; A_3: Picnic shelter; A_4: Lease to operator.

States: S_1: Busy season; S_2: Typical season; S_3: Quiet season.

$$\boldsymbol{q} = (0.25, 0.45, 0.30), \qquad 0.25 + 0.45 + 0.30 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P(A_i, S_j).$$

Multiply each state payoff by its probability, including any negative payoff, and add the weighted amounts. All EMVs are in thousand USD.

$$\begin{aligned} EMV(A_1) &= 0.25(117) + 0.45(43) + 0.30(-29) \\ &= 29.25 + 19.35 - 8.70 = 39.90, \\ EMV(A_2) &= 0.25(89) + 0.45(64) + 0.30(23) \\ &= 22.25 + 28.80 + 6.90 = 57.95, \\ EMV(A_3) &= 0.25(63) + 0.45(57) + 0.30(40) \\ &= 15.75 + 25.65 + 12 = 53.40, \\ EMV(A_4) &= 0.25(32) + 0.45(32) + 0.30(32) \\ &= 8 + 14.40 + 9.60 = 32. \end{aligned}$$

$$\max\{39.90, 57.95, 53.40, 32\} = 57.95.$$

Choose **Covered terrace** (A_2), with expected profit **57.95 thousand USD** (USD 57,950). EMV is a probability-weighted expected payoff, not a guaranteed outcome. The realised profit will be the payoff for the state that actually occurs.

11 Translation Capacity [C6-2]

A translation agency must choose a capacity plan before client orders are known. The possible states, in column order, are Order rush, Busy orders, Routine orders, Few orders. Profits are in thousand USD.

Alternative	S_1	S_2	S_3	S_4
$P(S_j)$	0.15	0.35	0.25	0.25
Permanent staff (A_1)	137	81	29	-43
Staff plus freelancers (A_2)	99	86	59	17
Core staff (A_3)	65	67	61	48

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - EMV / Bayes [2]

Alternatives: A_1: Permanent staff; A_2: Staff plus freelancers; A_3: Core staff.

States: S_1: Order rush; S_2: Busy orders; S_3: Routine orders; S_4: Few orders.

$$\boldsymbol{q} = (0.15, 0.35, 0.25, 0.25), \qquad 0.15 + 0.35 + 0.25 + 0.25 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P(A_i, S_j).$$

Multiply each state payoff by its probability, including any negative payoff, and add the weighted amounts. All EMVs are in thousand USD.

$$\begin{aligned} EMV(A_1) &= 0.15(137) + 0.35(81) + 0.25(29) + 0.25(-43) \\ &= 20.55 + 28.35 + 7.25 - 10.75 = 45.40, \\ EMV(A_2) &= 0.15(99) + 0.35(86) + 0.25(59) + 0.25(17) \\ &= 14.85 + 30.10 + 14.75 + 4.25 = 63.95, \\ EMV(A_3) &= 0.15(65) + 0.35(67) + 0.25(61) + 0.25(48) \\ &= 9.75 + 23.45 + 15.25 + 12 = 60.45. \end{aligned}$$

$$\max\{45.40, 63.95, 60.45\} = 63.95.$$

Choose **Staff plus freelancers** (A_2), with expected profit **63.95 thousand USD** (USD 63,950). EMV is a probability-weighted expected payoff, not a guaranteed outcome. The realised profit will be the payoff for the state that actually occurs.

12 Mobile Game Release [C7-3]

A game developer must choose a wide release or a limited release before player uptake is known. The possible states, in column order, are Viral uptake, Moderate uptake, Low uptake. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Wide release (A_1)	108	44	-10
Limited release (A_2)	58	64	30

Suppose $P(S_1) = p$ and $P(S_2) = p/2$; determine $P(S_3)$ from the sum-to-one condition.

Questions/Tasks

C7 State the preferred alternative for every probability value ($0 \le p \le 1$). **[3]**

C7 - Probability sensitivity analysis [3]

Alternatives: A_1: Wide release; A_2: Limited release.

States: S_1: Viral uptake; S_2: Moderate uptake; S_3: Low uptake. The mutually exclusive and exhaustive state probabilities sum to one:

$$P(S_3) = 1 - p - p/2 = 1 - 3p/2, \qquad \boldsymbol{q} = \left(p, \frac{p}{2}, 1 - \frac{3p}{2}\right).$$

Every probability must lie in $[0, 1]$:

$$0 \le p \le 1, \quad 0 \le p/2 \le 1 \iff 0 \le p \le 2, \quad 0 \le 1 - 3p/2 \le 1 \iff 0 \le p \le \frac{2}{3}.$$

Their intersection is $0 \le p \le \frac{2}{3}$. The remaining part of $[0, 1]$, $\frac{2}{3} < p \le 1$, is infeasible because $P(S_3) < 0$. No decision is valid there. Expected profits are in thousand USD:

$$\begin{aligned} E_1 &= EMV(A_1) = 108p + 44(p/2) - 10(1 - 3p/2) \\ &= 108p + 22p - 10 + 15p = -10 + 145p, \\ E_2 &= EMV(A_2) = 58p + 64(p/2) + 30(1 - 3p/2) \\ &= 58p + 32p + 30 - 45p = 30 + 45p. \end{aligned}$$

Equating the two linear EMVs gives

$$E_1 = E_2 \iff 100p = 40 \iff p = \frac{2}{5}.$$

This is the only threshold, since the slopes differ, and $0 < \frac{2}{5} < \frac{2}{3}$. At the tie, $E_1 = E_2 = 48$ thousand USD. The difference $E_1 - E_2 = -40 + 100p$ is negative below the threshold and positive above it. At $p = 0$, $(E_1, E_2) = (-10, 30)$; at $p = \frac{2}{3}$, $(E_1, E_2) = (\frac{260}{3}, 60)$. Both feasible endpoints agree with the rule:

$$\begin{cases} \text{Limited release } (A_2), & 0 \le p < \frac{2}{5}, \\ \text{Both alternatives tied}, & p = \frac{2}{5}, \\ \text{Wide release } (A_1), & \frac{2}{5} < p \le \frac{2}{3}. \end{cases}$$

Either alternative is optimal at the tie. Expected profit is not guaranteed profit.

13 Timber Supply Agreement [C7-3]

A timber merchant must choose spot sales or a reserved-supply agreement before next season's timber availability is known. The possible states, in column order, are Scarce timber, Balanced supply, Abundant timber. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Spot sales (A_1)	30	66	94
Reserved supply (A_2)	85	46	40

Suppose $P(S_1) = 2p$ and $P(S_2) = p$; determine $P(S_3)$ from the sum-to-one condition.

Questions/Tasks

C7 State the preferred alternative for every probability value ($0 \le p \le 1$). **[3]**

C7 - Probability sensitivity analysis [3]

Alternatives: A_1: Spot sales; A_2: Reserved supply.

States: S_1: Scarce timber; S_2: Balanced supply; S_3: Abundant timber. The mutually exclusive and exhaustive state probabilities sum to one:

$$P(S_3) = 1 - 2p - p = 1 - 3p, \qquad \boldsymbol{q} = (2p, p, 1 - 3p).$$

Every probability must lie in $[0, 1]$:

$$0 \le 2p \le 1 \iff 0 \le p \le \frac{1}{2}, \quad 0 \le p \le 1, \quad 0 \le 1 - 3p \le 1 \iff 0 \le p \le \frac{1}{3}.$$

Their intersection is $0 \le p \le \frac{1}{3}$. The remaining part of $[0, 1]$, $\frac{1}{3} < p \le 1$, is infeasible because $P(S_3) < 0$. No decision is valid there. For $p > \frac{1}{2}$, $P(S_1) > 1$ as well. Expected profits are in thousand USD:

$$\begin{aligned} E_1 &= EMV(A_1) = 30(2p) + 66p + 94(1 - 3p) \\ &= 60p + 66p + 94 - 282p = 94 - 156p, \\ E_2 &= EMV(A_2) = 85(2p) + 46p + 40(1 - 3p) \\ &= 170p + 46p + 40 - 120p = 40 + 96p. \end{aligned}$$

Equating the two linear EMVs gives

$$E_1 = E_2 \iff -252p = -54 \iff p = \frac{3}{14}.$$

This is the only threshold, since the slopes differ, and $0 < \frac{3}{14} < \frac{1}{3}$. At the tie, $E_1 = E_2 = \frac{424}{7}$ thousand USD. The difference $E_1 - E_2 = 54 - 252p$ is positive below the threshold and negative above it. At $p = 0$, $(E_1, E_2) = (94, 40)$; at $p = \frac{1}{3}$, $(E_1, E_2) = (42, 72)$. Both feasible endpoints agree with the rule:

$$\begin{cases} \text{Spot sales } (A_1), & 0 \le p < \frac{3}{14}, \\ \text{Both alternatives tied}, & p = \frac{3}{14}, \\ \text{Reserved supply } (A_2), & \frac{3}{14} < p \le \frac{1}{3}. \end{cases}$$

Either alternative is optimal at the tie. Expected profit is not guaranteed profit.

14 Summarized Solutions [C1–C7]

The numbered summaries follow the exam order. A_i and S_j use the original row and column order. $E_i = EMV(A_i)$; b_j is the best payoff in state S_j; $r_{ij} = b_j - P(A_i, S_j)$. Units are stated for each problem.

14.1 Community Science Sessions C1–3 C2–6

Alternatives: A_1: Astronomy session; A_2: Robotics session.

States: S_1: Busy demand; S_2: Quiet demand. Uncertain event: actual demand after the choice. Context: community science centre selecting a session; consequences are net contributions in USD.

$$P_{ij} = q_{ij}(v_i + d_j) + f_i.$$

The fixed sponsorship is added once; the two per-participant rates multiply attendance.

Payoff (USD)	S_1: busy	S_2: quiet
Astronomy (A_1)	$28(14+5)+90 = 622$	$17(14+3)+90 = 379$
Robotics (A_2)	$23(17+5)+65 = 571$	$14(17+3)+65 = 345$

Astronomy gives the higher contribution in both demand states.



14.2 Kayak Rentals C1–3 C3–6

Alternatives: A_1: Single kayaks; A_2: Tandem kayaks.

States: S_1: Busy demand; S_2: Quiet demand. Uncertain event: actual demand after the choice. Context: kayak rental business selecting a rental type; consequences are seasonal profits in USD.

$$P_{ij} = q_{ij}(b_i + s_j) - q_{ij}c_i - F_j.$$

Revenue includes the surcharge; the seasonal fixed cost is subtracted once.

Profit (USD)	S_1: busy	S_2: quiet
Single (A_1)	$84(36+7) - 84(20) - 540 = 1392$	$56(36+4) - 56(20) - 620 = 500$
Tandem (A_2)	$66(49+7) - 66(29) - 540 = 1242$	$38(49+4) - 38(29) - 620 = 292$

Single kayaks earn more in both demand states.



14.3 Bicycle Assembly C4–2

Alternatives: A_1: Large factory; A_2: Standard line; A_3: Compact line; A_4: Licensed production.
States: S_1: Boom; S_2: Stable market; S_3: Recession.

Alternative	Row maximum M_i	Row minimum m_i
Large factory (A_1)	132	-27
Standard line (A_2)	91	19
Compact line (A_3)	61	33
Licensed production (A_4)	26	26

$$\max_i M_i = \max\{132, 91, 61, 26\} = 132, \qquad \max_i m_i = \max\{-27, 19, 33, 26\} = 33.$$

Maximax: choose **Large factory** (A_1), with best-case profit 132 thousand USD.
Maximin: choose **Compact line** (A_3), with worst-case profit 33 thousand USD.



14.4 Solar Installation C4–2

Alternatives: A_1: Regional rollout; A_2: Town rollout; A_3: Village rollout; A_4: Service agreement.
States: S_1: Rapid adoption; S_2: Moderate adoption; S_3: Slow adoption.

Alternative	Row maximum M_i	Row minimum m_i
Regional rollout (A_1)	168	-39
Town rollout (A_2)	108	17
Village rollout (A_3)	73	31
Service agreement (A_4)	35	35

$$\max_i M_i = \max\{168, 108, 73, 35\} = 168, \qquad \max_i m_i = \max\{-39, 17, 31, 35\} = 35.$$

Maximax: choose **Regional rollout** (A_1), with best-case profit 168 thousand USD.
Maximin: choose **Service agreement** (A_4), with worst-case profit 35 thousand USD.



14.5 Equipment Leasing C4–2

Alternatives: A_1: Buy large fleet; A_2: Buy medium fleet; A_3: Lease small fleet; A_4: Pool equipment.
States: S_1: Peak utilisation; S_2: Busy utilisation; S_3: Routine utilisation; S_4: Quiet utilisation.

Alternative	Row maximum M_i	Row minimum m_i
Buy large fleet (A_1)	156	-46
Buy medium fleet (A_2)	113	11
Lease small fleet (A_3)	77	25
Pool equipment (A_4)	54	32

$$\max_i M_i = \max\{156, 113, 77, 54\} = 156, \qquad \max_i m_i = \max\{-46, 11, 25, 32\} = 32.$$

Maximax: choose **Buy large fleet** (A_1), with best-case profit 156 thousand USD.
Maximin: choose **Pool equipment** (A_4), with worst-case profit 32 thousand USD.



14.6 Festival Meals C5–2

Alternatives: A_1: Full menu; A_2: Balanced menu; A_3: Limited menu.
States: S_1: Crowded; S_2: Average attendance; S_3: Sparse attendance.

$$\boldsymbol{b} = \big(87, 54, 34\big), \qquad b_j = \max_k P(A_k, S_j).$$

All payoffs and regrets in this calculation are in thousand USD.

Alternative	S_1	S_2	S_3	Maximum regret
A_1	$87-87$ $=0$	$54-37$ $=17$	$34-(-16)$ $=50$	50
A_2	$87-65$ $=22$	$54-54$ $=0$	$34-21$ $=13$	22
A_3	$87-43$ $=44$	$54-41$ $=13$	$34-34$ $=0$	44

$$\min_i \max_j r(A_i, S_j) = \min\{50, 22, 44\} = 22.$$

Choose **Balanced menu** (A_2): its maximum regret is **22 thousand USD**.



14.7 Laundry Capacity C5–2

Alternatives: A_1: Build plant; A_2: Upgrade machines; A_3: Lease machines; A_4: Contract service.
States: S_1: Heavy workload; S_2: Routine workload; S_3: Light workload.

$$\boldsymbol{b} = \big(126, 73, 41\big), \qquad b_j = \max_k P(A_k, S_j).$$

All payoffs and regrets in this calculation are in thousand USD.

Alternative	S_1	S_2	S_3	Maximum regret
A_1	$126-126$ $=0$	$73-47$ $=26$	$41-(-32)$ $=73$	73
A_2	$126-94$ $=32$	$73-73$ $=0$	$41-19$ $=22$	32
A_3	$126-69$ $=57$	$73-61$ $=12$	$41-38$ $=3$	57
A_4	$126-41$ $=85$	$73-41$ $=32$	$41-41$ $=0$	85

$$\min_i \max_j r(A_i, S_j) = \min\{73, 32, 57, 85\} = 32.$$

Choose **Upgrade machines** (A_2): its maximum regret is **32 thousand USD**.



14.8 Juice Production C5–2

Alternatives: A_1: Large batch; A_2: Staged batches; A_3: Small batch.

States: S_1: Exceptional sales; S_2: Brisk sales; S_3: Ordinary sales; S_4: Poor sales.

$$\boldsymbol{b} = \big(1080, 730, 510, 390\big), \qquad b_j = \max_k P(A_k, S_j).$$

All payoffs and regrets in this calculation are in USD.

Alternative	S_1	S_2	S_3	S_4	Maximum regret
A_1	$1080 - 1080 = 0$	$730 - 610 = 120$	$510 - 230 = 280$	$390 - (-270) = 660$	660
A_2	$1080 - 810 = 270$	$730 - 730 = 0$	$510 - 510 = 0$	$390 - 150 = 240$	270
A_3	$1080 - 530 = 550$	$730 - 550 = 180$	$510 - 480 = 30$	$390 - 390 = 0$	550

$$\min_i \max_j r(A_i, S_j) = \min\{660, 270, 550\} = 270.$$

Choose **Staged batches** (A_2): its maximum regret is **270 USD**.



14.9 Coffee Roasting C6–2

Alternatives: A_1: Specialty roast; A_2: Blended roast; A_3: Everyday roast.
States: S_1: Export boom; S_2: Regular exports; S_3: Export slump.

$$\boldsymbol{q} = (0.35, 0.40, 0.25), \qquad 0.35 + 0.40 + 0.25 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$\begin{aligned} EMV(A_1) &= 0.35(98) + 0.40(49) + 0.25(-17) = 49.65, \\ EMV(A_2) &= 0.35(74) + 0.40(63) + 0.25(27) = 57.85, \\ EMV(A_3) &= 0.35(51) + 0.40(53) + 0.25(42) = 49.55. \end{aligned}$$

$$\max\{49.65, 57.85, 49.55\} = 57.85.$$

Choose **Blended roast** (A_2), with expected profit **57.85 thousand USD** (USD 57,850). Expected profit is not guaranteed profit.



14.10 Farm Visitor Facilities C6–2

Alternatives: A_1: Visitor centre; A_2: Covered terrace; A_3: Picnic shelter; A_4: Lease to operator.
States: S_1: Busy season; S_2: Typical season; S_3: Quiet season.

$$\boldsymbol{q} = (0.25, 0.45, 0.30), \qquad 0.25 + 0.45 + 0.30 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$\begin{aligned} EMV(A_1) &= 0.25(117) + 0.45(43) + 0.30(-29) = 39.90, \\ EMV(A_2) &= 0.25(89) + 0.45(64) + 0.30(23) = 57.95, \\ EMV(A_3) &= 0.25(63) + 0.45(57) + 0.30(40) = 53.40, \\ EMV(A_4) &= 0.25(32) + 0.45(32) + 0.30(32) = 32. \end{aligned}$$

$$\max\{39.90, 57.95, 53.40, 32\} = 57.95.$$

Choose **Covered terrace** (A_2), with expected profit **57.95 thousand USD** (USD 57,950). Expected profit is not guaranteed profit.



14.11 Translation Capacity C6–2

Alternatives: A_1: Permanent staff; A_2: Staff plus freelancers; A_3: Core staff.
States: S_1: Order rush; S_2: Busy orders; S_3: Routine orders; S_4: Few orders.

$$\boldsymbol{q} = (0.15, 0.35, 0.25, 0.25), \qquad 0.15 + 0.35 + 0.25 + 0.25 = 1.$$

All probabilities lie in $[0, 1]$, so they are nonnegative and valid.

$$\begin{aligned} EMV(A_1) &= 0.15(137) + 0.35(81) + 0.25(29) + 0.25(-43) = 45.40, \\ EMV(A_2) &= 0.15(99) + 0.35(86) + 0.25(59) + 0.25(17) = 63.95, \\ EMV(A_3) &= 0.15(65) + 0.35(67) + 0.25(61) + 0.25(48) = 60.45. \end{aligned}$$

$$\max\{45.40, 63.95, 60.45\} = 63.95.$$

Choose **Staff plus freelancers** (A_2), with expected profit **63.95 thousand USD** (USD 63,950). Expected profit is not guaranteed profit.



14.12 Mobile Game Release C7–3

Alternatives: A_1: Wide release; A_2: Limited release.

States: S_1: Viral uptake; S_2: Moderate uptake; S_3: Low uptake. The mutually exclusive and exhaustive state probabilities sum to one:

$$P(S_3) = 1 - p - p/2 = 1 - 3p/2, \qquad \boldsymbol{q} = \left(p, \frac{p}{2}, 1 - \frac{3p}{2}\right).$$

Every probability must lie in $[0, 1]$:

$$0 \le p \le 1, \quad 0 \le p/2 \le 1 \iff 0 \le p \le 2, \quad 0 \le 1 - 3p/2 \le 1 \iff 0 \le p \le \frac{2}{3}.$$

Their intersection is $\boxed{0 \le p \le \frac{2}{3}}$. The remaining part of $[0, 1]$, $\frac{2}{3} < p \le 1$, is infeasible because $P(S_3) < 0$. No decision is valid there. Expected profits are in thousand USD:

$$\begin{aligned} E_1 &= EMV(A_1) = 108p + 44(p/2) - 10(1 - 3p/2) \\ &= 108p + 22p - 10 + 15p = -10 + 145p, \\ E_2 &= EMV(A_2) = 58p + 64(p/2) + 30(1 - 3p/2) \\ &= 58p + 32p + 30 - 45p = 30 + 45p. \end{aligned}$$

Equating the two linear EMVs gives

$$E_1 = E_2 \iff 100p = 40 \iff \boxed{p = \frac{2}{5}}.$$

This is the only threshold, since the slopes differ, and $0 < \frac{2}{5} < \frac{2}{3}$. At the tie, $E_1 = E_2 = 48$ thousand USD. The difference $E_1 - E_2 = -40 + 100p$ is negative below the threshold and positive above it. At $p = 0$, $(E_1, E_2) = (-10, 30)$; at $p = \frac{2}{3}$, $(E_1, E_2) = (\frac{260}{3}, 60)$. Both feasible endpoints agree with the rule:

$$\begin{cases} \text{Limited release } (A_2), & 0 \le p < \frac{2}{5}, \\ \text{Both alternatives tied}, & p = \frac{2}{5}, \\ \text{Wide release } (A_1), & \frac{2}{5} < p \le \frac{2}{3}. \end{cases}$$

Either alternative is optimal at the tie. Expected profit is not guaranteed profit.



14.13 Timber Supply Agreement C7–3

Alternatives: A_1: Spot sales; A_2: Reserved supply.

States: S_1: Scarce timber; S_2: Balanced supply; S_3: Abundant timber. The mutually exclusive and exhaustive state probabilities sum to one:

$$P(S_3) = 1 - 2p - p = 1 - 3p, \qquad \boldsymbol{q} = (2p, p, 1 - 3p).$$

Every probability must lie in $[0, 1]$:

$$0 \le 2p \le 1 \iff 0 \le p \le \frac{1}{2}, \quad 0 \le p \le 1, \quad 0 \le 1 - 3p \le 1 \iff 0 \le p \le \frac{1}{3}.$$

Their intersection is $\boxed{0 \le p \le \frac{1}{3}}$. The remaining part of $[0, 1]$, $\frac{1}{3} < p \le 1$, is infeasible because $P(S_3) < 0$. No decision is valid there. For $p > \frac{1}{2}$, $P(S_1) > 1$ as well. Expected profits are in thousand USD:

$$\begin{aligned} E_1 &= EMV(A_1) = 30(2p) + 66p + 94(1 - 3p) \\ &= 60p + 66p + 94 - 282p = 94 - 156p, \\ E_2 &= EMV(A_2) = 85(2p) + 46p + 40(1 - 3p) \\ &= 170p + 46p + 40 - 120p = 40 + 96p. \end{aligned}$$

Equating the two linear EMVs gives

$$E_1 = E_2 \iff -252p = -54 \iff \boxed{p = \frac{3}{14}}.$$

This is the only threshold, since the slopes differ, and $0 < \frac{3}{14} < \frac{1}{3}$. At the tie, $E_1 = E_2 = \frac{424}{7}$ thousand USD. The difference $E_1 - E_2 = 54 - 252p$ is positive below the threshold and negative above it. At $p = 0$, $(E_1, E_2) = (94, 40)$; at $p = \frac{1}{3}$, $(E_1, E_2) = (42, 72)$. Both feasible endpoints agree with the rule:

$$\begin{cases} \text{Spot sales } (A_1), & 0 \le p < \frac{3}{14}, \\ \text{Both alternatives tied}, & p = \frac{3}{14}, \\ \text{Reserved supply } (A_2), & \frac{3}{14} < p \le \frac{1}{3}. \end{cases}$$

Either alternative is optimal at the tie. Expected profit is not guaranteed profit.

